

TECHSHIP INTERNSHIP

Day 3- Functions,Arrays,Objects & ES6 Concepts

Submitted By : Kelna Sebastian

Date : 02-06-2026

During Day 3 of the internship, I learned how to work with functions, arrays, objects, and modern JavaScript ES6 features. I gained a better understanding of how reusable functions help organize code and how arrays and objects can be used to store and manage structured data efficiently. I also explored array methods such as `forEach()`, `filter()`, `map()`, and `reduce()`, which are useful for processing and manipulating data in real-world applications.

The ES6 feature I found most useful was the spread operator because it allows objects and arrays to be copied and modified easily while keeping the original data unchanged. I also found template literals very helpful for generating clean and readable reports.

The main challenge I faced was understanding the `reduce()` method, particularly how its parameters work and the purpose of the initial value. While implementing the Student Management System assignment, I initially found it difficult to calculate averages and identify the top performer using array methods. However, after practicing with examples and debugging the code, I became more confident in using these concepts. Overall, this session improved my understanding of structured JavaScript programming, strengthened my problem-solving skills, and gave me hands-on experience in developing programs that resemble real-world data management systems.

The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a folder structure with 'Day1', 'Day2', and 'Day3'. Under 'Day3', there is an 'Exercise' folder containing 'Ex1.js' and a 'README.md' file. The main editor window displays the code in 'Ex1.js', which defines four functions: 'add', 'subtract', 'multiply', and 'divide'. Each function takes two arguments, 'a' and 'b', and returns the result of the corresponding operation. Below the function definitions, there are four console.log statements that call each function with specific values. The bottom panel shows the 'TERMINAL' tab with the command 'node Ex1.js' executed, resulting in the following output: 'Addition: 150', 'Subtraction: 6', 'Multiplication: 0', and 'Division: 0.5'.

```
1 function add(a, b) {
2   return a + b;
3 }
4
5 function subtract(a, b) {
6   return a - b;
7 }
8
9 function multiply(a, b) {
10  return a * b;
11 }
12
13 function divide(a, b) {
14   return a / b;
15 }
16
17 console.log("Addition:", add(50, 100));
18 console.log("Subtraction:", subtract(12, 6));
19 console.log("Multiplication:", multiply(0, 3));
20 console.log("Division:", divide(2, 4));
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex1.js
Addition: 150
Subtraction: 6
Multiplication: 0
Division: 0.5
D:\TECHSHIP\Day3\Exercise>

Ex1:Function Based Calculator

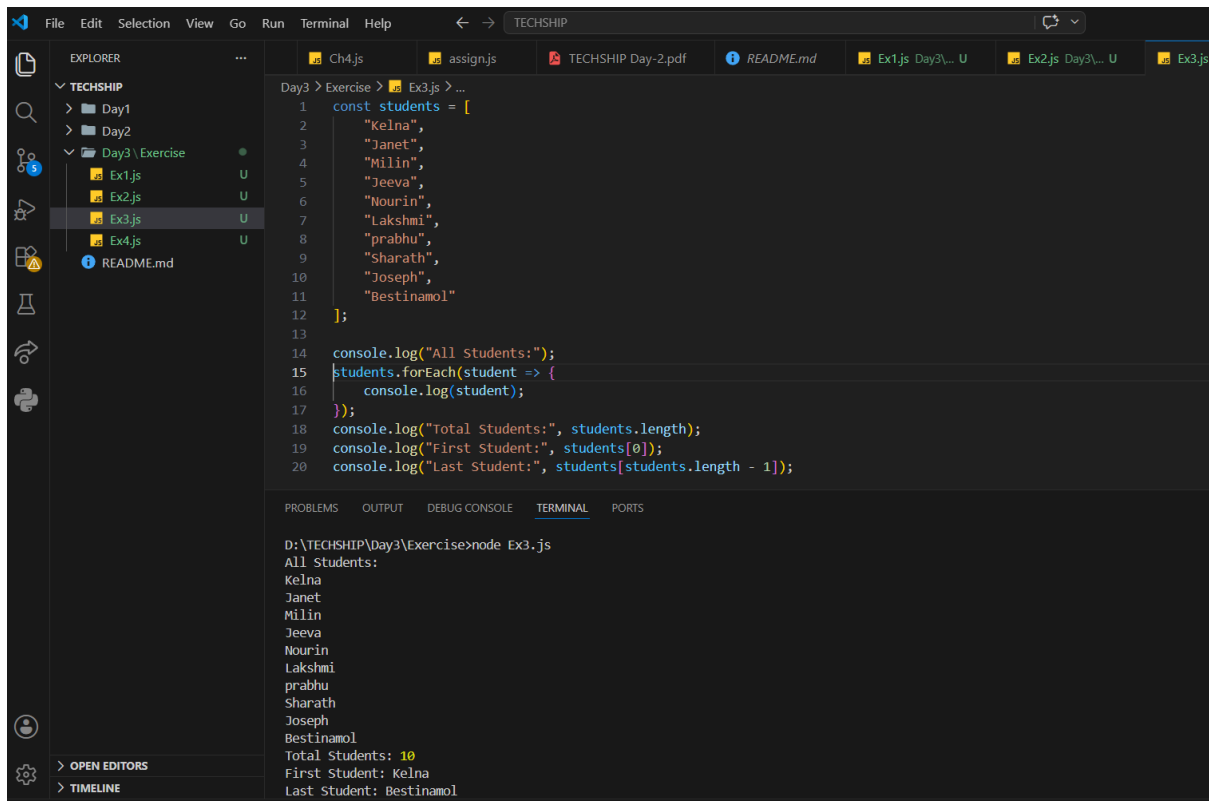
The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar shows the 'Ex2.js' file selected under the 'Day3' folder. The main editor window displays the code in 'Ex2.js', which defines a 'greetStudent' function. The function takes two arguments, 'name' and 'course', and uses console.log to print a welcome message and the course name. Below the function definition, there is a call to 'greetStudent' with the values 'Kelna' and 'MBA'. The bottom panel shows the 'TERMINAL' tab with the command 'node Ex2.js' executed, resulting in the following output: 'Welcome Kelna!' and 'You are enrolled in MBA.'.

```
1 function greetStudent(name, course) {
2   console.log(`Welcome ${name}!`);
3   console.log(`You are enrolled in ${course}.`);
4 }
5
6 greetStudent("Kelna", "MBA");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Exercise>node Ex2.js
Welcome Kelna!
You are enrolled in MBA.
D:\TECHSHIP\Day3\Exercise>

Ex2:Student Greeting Function



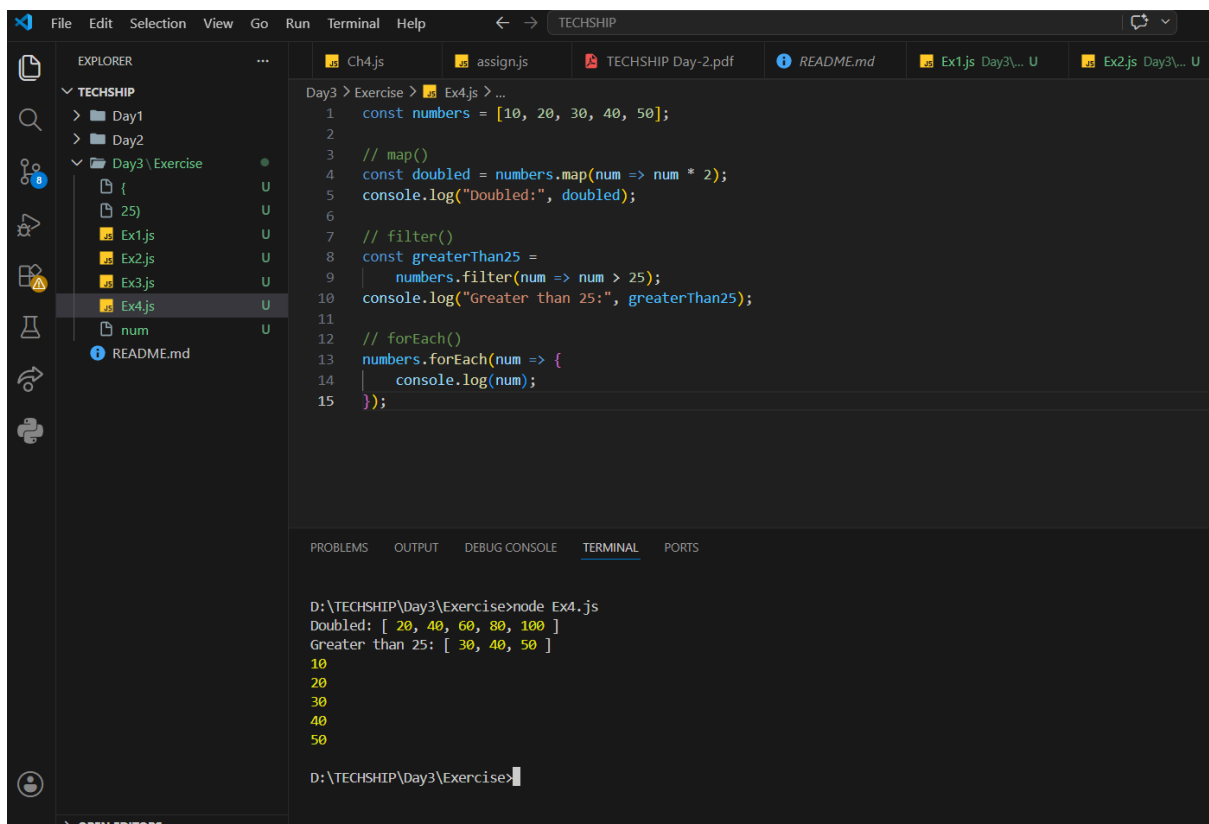
The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows the file structure: Day1, Day2, and Day3. Under Day3, there is an 'Exercise' folder containing Ex1.js, Ex2.js, Ex3.js (selected), and Ex4.js. The main editor area displays the code in Ex3.js:

```
Day3 > Exercise > Ex3.js > ...
1  const students = [
2      "Kelna",
3      "Janet",
4      "Milin",
5      "Jeeva",
6      "Nourin",
7      "Lakshmi",
8      "prabhu",
9      "Sharath",
10     "Joseph",
11     "Bestinamol"
12 ];
13
14 console.log("All Students:");
15 students.forEach(student => {
16     console.log(student);
17 });
18 console.log("Total Students:", students.length);
19 console.log("First Student:", students[0]);
20 console.log("Last Student:", students[students.length - 1]);
```

The terminal at the bottom shows the output of running 'node Ex3.js' in the directory 'D:\TECHSHIP\Day3\Exercise':

```
D:\TECHSHIP\Day3\Exercise>node Ex3.js
All Students:
Kelna
Janet
Milin
Jeeva
Nourin
Lakshmi
prabhu
Sharath
Joseph
Bestinamol
Total Students: 10
First Student: Kelna
Last Student: Bestinamol
```

Ex3:Working with Arrays



The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar shows the 'Exercise' folder with Ex4.js selected. The main editor area displays the code in Ex4.js:

```
Day3 > Exercise > Ex4.js > ...
1  const numbers = [10, 20, 30, 40, 50];
2
3  // map()
4  const doubled = numbers.map(num => num * 2);
5  console.log("Doubled:", doubled);
6
7  // filter()
8  const greaterThan25 =
9      numbers.filter(num => num > 25);
10 console.log("Greater than 25:", greaterThan25);
11
12 // forEach()
13 numbers.forEach(num => {
14     console.log(num);
15 });
```

The terminal at the bottom shows the output of running 'node Ex4.js' in the directory 'D:\TECHSHIP\Day3\Exercise':

```
D:\TECHSHIP\Day3\Exercise>node Ex4.js
Doubled: [ 20, 40, 60, 80, 100 ]
Greater than 25: [ 30, 40, 50 ]
10
20
30
40
50

D:\TECHSHIP\Day3\Exercise>
```

Ex4:Array Methods

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure under 'TECHSHIP'. The file 'Ex5.js' is selected in the 'Day3 \ Exercise' folder. The main editor displays the code in 'Ex5.js':

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript, Flutter"]}
7 };
8
9 console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
D:\TECHSHIP\Day3\Exercise>node Ex5.js
Kelna
21
AISAT College
BTech
[ 'HTML', 'CSS', 'JavaScript, Flutter' ]
D:\TECHSHIP\Day3\Exercise>
```

Ex5:Student Object

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure under 'TECHSHIP'. The file 'Ex5.js' is selected in the 'Day3 \ Exercise' folder. The main editor displays the code in 'Ex5.js':

```
1 const student = {
2   name: "Kelna",
3   age: 21,
4   college: "AISAT College",
5   course: "BTech",
6   skills: ["HTML", "CSS", "JavaScript", "Flutter"]}
7 };
8
9 console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
D:\TECHSHIP\Day3\Exercise>node Ex5.js
Kelna
21
AISAT College
BTech
[ 'HTML', 'CSS', 'JavaScript', 'Flutter' ]
D:\TECHSHIP\Day3\Exercise>
```

Ex6:Employee Object

The screenshot shows a VS Code editor with a project named 'TECHSHIP'. The Explorer sidebar shows a folder structure with 'Day1', 'Day2', and 'Day3'. Inside 'Day3', there is an 'Exercise' folder containing files 'Ex1.js' through 'Ex6.js'. The file 'Ex6.js' is selected and open in the editor. The code in 'Ex6.js' defines an employee object with the following properties: `employeeId: 102`, `name: "Aadhithya"`, `department: "HR"`, and `salary: 30000`. It then logs the object before and after updating the salary to 40000. The terminal at the bottom shows the command `D:\TECHSHIP\Day3\Exercise>node Ex6.js` and its output: `Before Update: { employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 30000 } After Update: { employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 40000 }`.

```
1 const employee = {
2   employeeId: 102,
3   name: "Aadhithya",
4   department: "HR",
5   salary: 30000
6 };
7
8 console.log("Before Update:");
9 console.log(employee);
10
11 // Update salary
12 employee.salary = 40000;
13
14 console.log("After Update:");
15 console.log(employee);
```

D:\TECHSHIP\Day3\Exercise>node Ex6.js
Before Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 30000 }
After Update:
{ employeeId: 102, name: 'Aadhithya', department: 'HR', salary: 40000 }

D:\TECHSHIP\Day3\Exercise>

Ex7:ES6 Practise

Challenge Problems

The screenshot shows a VS Code editor with a project named 'TECHSHIP'. The Explorer sidebar shows a folder structure with 'Day1', 'Day2', and 'Day3'. Inside 'Day3', there is an 'Exercise' folder containing files 'Ex1.js' through 'Ex7.js'. The file 'Ex7.js' is selected and open in the editor. The code in 'Ex7.js' demonstrates ES6 features: template literals, destructuring, and the spread operator. It creates a student object, logs it, and then creates a new student object with the same properties plus a 'college' property. The terminal at the bottom shows the command `D:\TECHSHIP\Day3\Exercise>node Ex7.js` and its output: `Hi, I am Kelna. I am studying BTech CSE. Kelna BTech CSE { name: 'kelna', course: 'BTech CSE', age: 21, college: 'xyz college' }`.

```
1 const student = {
2   name: 'kelna',
3   course: 'BTech CSE',
4   age: 21
5 };
6 //Template Literal
7 console.log(
8   `Hi, I am ${student.name}.
9   I am studying ${student.course}.`
10 );
11 //Destructuring
12 const { name, course } = student;
13 console.log(name);
14 console.log(course);
15 //Spread Operator
16 const newStudent = {
17   ...student,
18   college: "xyz college"
19 };
20 console.log(newStudent);
```

D:\TECHSHIP\Day3\Exercise>node Ex7.js
Hi, I am Kelna.
I am studying BTech CSE.
Kelna
BTech CSE
{ name: 'kelna', course: 'BTech CSE', age: 21, college: 'xyz college' }

D:\TECHSHIP\Day3\Exercise>

1.Student Marks Analyser

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
├── Day1
├── Day2
│   ├── Challenges
│   └── Exercise
│       ├── assign.js
│       └── TECHSHIP Day-2.pdf
├── Day3
│   ├── Challenge
│   │   ├── Ch1.js
│   │   └── Exercise
│   │       ├── Ex1.js
│   │       ├── Ex2.js
│   │       ├── Ex3.js
│   │       ├── Ex4.js
│   │       ├── Ex5.js
│   │       ├── Ex6.js
│   │       └── Ex7.js
│   └── README.md
└── ...

Day3 > Challenge > Ch1.js > marks
1 const marks = [90, 40, 86, 65, 78];
2
3 // Total
4 const total = marks.reduce((sum, mark) => sum + mark, 0);
5
6 // Highest
7 const highest = Math.max(...marks);
8
9 // Lowest
10 const lowest = Math.min(...marks);
11
12 // Average
13 const average = total / marks.length;
14
15 console.log("Total Marks:", total);
16 console.log("Highest Mark:", highest);
17 console.log("Lowest Mark:", lowest);
18 console.log("Average Mark:", average);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Challenge>node Ch1.js
Total Marks: 359
Highest Mark: 90
Lowest Mark: 40
Average Mark: 71.8

D:\TECHSHIP\Day3\Challenge>
```

2.Product Filter System

```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
├── Day1
├── Day2
│   ├── Challenges
│   └── Exercise
│       ├── assign.js
│       └── TECHSHIP Day-2.pdf
├── Day3
│   ├── Challenge
│   │   ├── Ch1.js
│   │   ├── Ch2.js
│   │   └── Exercise
│   │       ├── Ex1.js
│   │       ├── Ex2.js
│   │       ├── Ex3.js
│   │       ├── Ex4.js
│   │       ├── Ex5.js
│   │       ├── Ex6.js
│   │       └── Ex7.js
│   └── README.md
└── ...

Day3 > Challenge > Ch2.js > expensiveProducts > products.filter() callback
1 const products = [
2   { name: "Mouse", price: 700 },
3   { name: "Keyboard", price: 5000 },
4   { name: "Monitor", price: 20000 },
5   { name: "Speaker", price: 5000 }
6 ];
7
8 const expensiveProducts =
9   products.filter(product => product.price > 1000);
10
11 console.log(expensiveProducts);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

D:\TECHSHIP\Day3\Challenge>node Ch2.js
[
  { name: 'Keyboard', price: 5000 },
  { name: 'Monitor', price: 20000 },
  { name: 'Speaker', price: 5000 }
]
```

3.Employee Directory

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure. The main editor area shows the code for `Ch3.js` in the `Day3 > Challenge` directory. The code defines an array of employee objects and uses `forEach` to log their names, and `filter` to log employees in the IT department. The terminal at the bottom shows the command `node Ch3.js` being executed, resulting in the names of the three employees and a filtered array of IT department employees.

```
Day3 > Challenge > Ch3.js > ...
1  const employees = [
2      { id: 1,
3        name: "Renu",
4        department: "IT"},
5      { id: 2,
6        name: "Jithu",
7        department: "HR"},
8      {id: 3,
9        name: "Geetha",
10       department: "IT"}];
11 // All employee names
12 employees.forEach(employee => {
13     console.log(employee.name);
14 });
15 // IT Department employees
16 const itEmployees =
17     employees.filter(
18         employee => employee.department === "IT");
19 console.log(itEmployees);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D:\TECHSHIP\Day3\Challenge>node Ch3.js
Renu
Jithu
Geetha
[
  { id: 1, name: 'Renu', department: 'IT' },
  { id: 3, name: 'Geetha', department: 'IT' }
]

D:\TECHSHIP\Day3\Challenge>
```

4.Course Enrollment System

Assignment

```
Ex6.js Day3... U  Ex7.js Day3... U  Ch1.js ...\\Challenge U  Ch2.js ...\\Challenge U  Ch3.js ...\\Challenge U  Ch4.js ...\\Challenge U
Day3 > assign.js > studentAverages > students.map() callback
1  const students = [{id: 100,
2      name: "Evelin",
3      age: 20,
4      course: "Cardiology Technician",
5      marks: [75, 80, 70, 85, 90]}, {
6      id: 101,
7      name: "Ann",
8      age: 21,
9      course: "BTech EC",
10     marks: [79, 82, 75, 88, 80]}, {
11     id: 102,
12     name: "Janet",
13     age: 21,
14     course: "BTech CS",
15     marks: [90, 95, 92, 88, 91]}, {
16     id: 103,
17     name: "Keina",
18     age: 22,
19     course: "BTech AI/ML",
20     marks: [65, 70, 68, 72, 66]},
21     { id: 104,
22         name: "Divya",
23         age: 21,
24         course: "Bsc Nurse",
25         marks: [78, 80, 75, 82, 79]}];
26     // Calculate average for each student
27     const studentAverages = students.map(student => {
28     const total = student.marks.reduce(
29         (sum, mark) => sum + mark,
30         0);
31     const average = total / student.marks.length;
32     return {
33         ...student,
34         average: average};});
35     // Students scoring above 75%
36     const topStudents = studentAverages.filter(
37         student => student.average > 75);
38     // Find top performer
39     const topper = studentAverages.reduce(
40         (highest, student) =>
41             student.average > highest.average
42             ? student
43             : highest);
44     console.log("=====");
45     console.log("STUDENT MANAGEMENT REPORT");
46     console.log("=====");
47     // Display all students with report
48     studentAverages.forEach(student => {let status;
49     if (student.average >= 80) {
50         status = "Excellent";
51     } else if (student.average >= 70) {
52         status = "Good";
53     } else {
54         status = "Needs Improvement";}
55     console.log(`ID: ${student.id}
56     Name: ${student.name}
57     Course: ${student.course}
58     Average Marks: ${student.average.toFixed(2)}
59     Status: ${status}`);});
60     // Students above 75%
61     console.log("=====");
62     console.log("Students Scoring Above 75%:");
63     topStudents.forEach(student => {console.log(`${student.name} - ${student.average.toFixed(2)} `)});
64     // Top performer
65     console.log("=====");
66     console.log(`Top Performer: ${topper.name}`);
67     console.log(`Highest Average: ${topper.average.toFixed(2)} `);
68     console.log("=====");
```

Student Management System

